

Accessible Colors

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How To Design Color Blind Friendly?

Accessible Colors

What is color blindness?

Color blindness—also known as color vision deficiency (CVD)—is a deficiency in distinguishing between different colors. It occurs when light-sensitive tissue in the back of the eye—the retina—fails to properly respond to variations in wavelengths of light, which enable people to see different colors.

Although color blindness is primarily an inherited condition, it can also result from cataracts or trauma to the eye, as well as a number of diseases, including Parkinson's, Kallman's Syndrome and diabetes. Damage to the retina caused by aging can additionally cause color blindness.

The term red-green color blindness is often used but actually not quite correct. Every type of color vision deficiency affects the whole color spectrum and therefore can not be reduced to just certain colors. Of course, for the most often encountered types Protanomaly (the milder form or red-weakness) or Protanopia (missing red cones, red-blindness) and Deuteranomaly (green-weakness) or Deuteranopia (green-blindness) the colors red and green are the most problematic and misinterpreted or indistinguishable ones. But people suffering from any of those forms have issues with the red or green part in every color, so the whole spectrum of colors is affected.

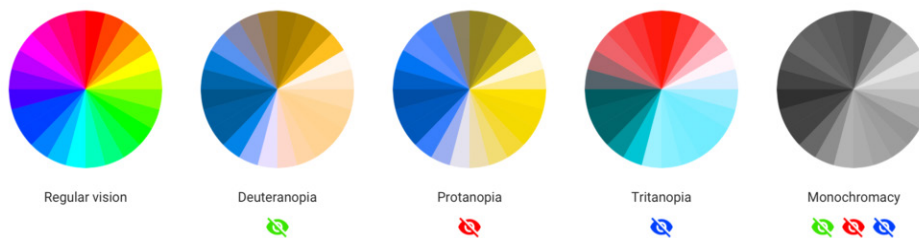
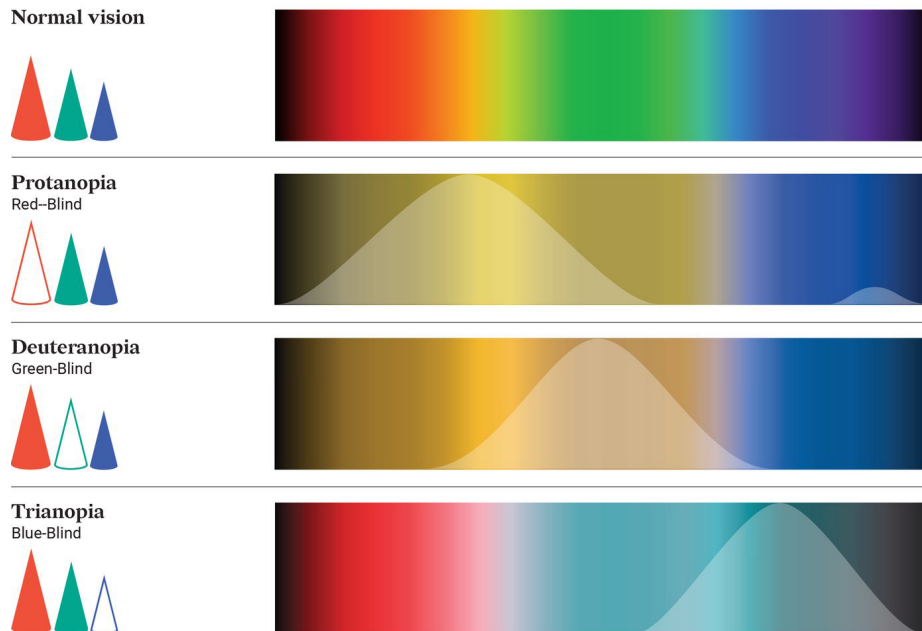
Besides red-green color blindness, which is encoded on the x-chromosome (sex-linked) and therefore much more common for men, there are also forms of color vision deficiency which are evenly distributed between male and female like Tritanomaly (blue-weakness) or Tritanopia (blue-blindness) and the real color blindness Achromatopsia (monochromacy) or the so called blue-cone monochromacy (only blue cones).

What colors can color blind people see?

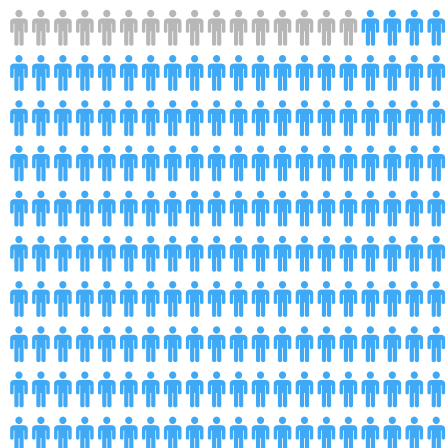
Actually, it depends on what type of cones don't work. There are three common types: protanopia (red-blind), deuteranopia (green-blind), and tritanopia (blue-blind). There can also be an intermediate state, color weakness - when the colors can be distinguished but some of the hue details are lost. There's also a very rare state when all the cones are not working and the person is completely color blind - it's called complete achromatopsia. In the next picture, the simulation of what colors can color blind people see is presented.

There are around 300 million people in the world who are colorblind. About 8% of men and 0.5% of women are colorblind, so making your chart color blind safe is a reasonable thing to do.

If you are wondering whether the colorblind minority of the world should factor into your design process, it is important to remember that 60% of those who are colorblind are severely impacted by their visual limitations, making it difficult for them to navigate and understand the world around them through design that rests solely on the foundations of colors.



16 of 200 men are color blind



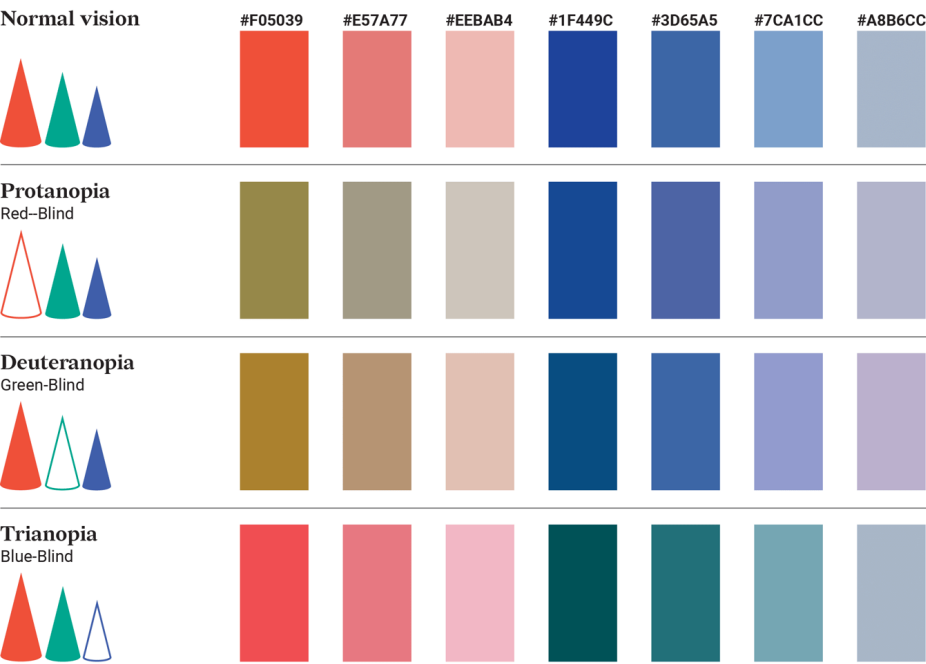
1 of 200 women are color blind



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Color blind palette

The picture below shows that for all of the types of color blindness there is a pair of colors that can be distinguished – red and blue – these are color blind safe colors. As for the green, it is the number one among the colors to avoid for the color blind. Most of the issues for colorblind people come from the prevalence of red-and-green combinations as a pair of opposites in design. The first rule of making a palette for colorblind – avoid combining red and green. So if you’re aiming to create a color blind-friendly palette try to use only two basic hues: blue and red (orange and yellow will also fit). The other colors should be made out of these two hues. All the variations can be made by using different saturation or lightness of the basic color. Based on these rules we’ve created a color blind palette and checked it for three color blindness types.



The palette worked for all types of color blindness, but you should always be aware that the variation of color-blind colors can vary significantly from person to person, so it’s always important to double-check. If you have a friend who is colorblind – ask for feedback. Alternatively, you can look up a colorblind community online and reach out to them. Don’t forget about people with complete color blindness and remember that 100% best colors for colorblind are black and white. So if your visualization works in greyscale you should be safe. Another advantage of using black and white data visualization is that they’re print-ready. But remember, if the palette is greyscale one should use other tools to distinguish chart elements.

How do you choose accessible colors?

There are a few factors to keep in mind when choosing accessible colors. First and foremost, consider contrast. You'll want to choose colors that provide maximum contrast. The best example of this is black text on a white background.

As a rule of thumb, you'll want to pair dark colored text with a light background or light colored text with a dark background. The WCAG 2.1 requires, "The visual presentation of text and images of text [to have] a contrast ratio of at least 4.5:1."

You'll also want to keep in mind that about 5% of people worldwide have color vision deficiency (color blindness). The most common form of color vision deficiency is red/green, but blue/yellow can also occur. You can still combine these colors in a palette, but you should try and avoid superimposing them.

How do you know if a color is accessible?

To determine if a color is accessible, you can check its contrast ratio against the colors you're pairing it with. There are several free online tools that can help you do this. Alternatively, if you want assurances the colors you choose will be accessible, try using Venngage Color Generator: venngage.com/tools/accessible-color-palette-generator#colorGenerator

How do you make an accessible color scheme?

Creating an accessible color palette starts with having empathy for your audience. Make sure you're well-versed in the different types of visual impairments and how colors affect the legibility of your content for individuals with these conditions.

Before you start to build your palette, it's also worth familiarizing yourself with the WCAG 2.1. Specifically, the section on contrast outlines rules for acceptable ratios. Your text needs to have a contrast ratio of at least 4.5:1.

Why is color accessibility important?

Designing with accessible colors is important because it ensures people with visual impairments can understand your content. It removes barriers and provides an equal opportunity for everyone — regardless of circumstances — to engage with your designs.

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Venngage Color Palette Generator



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Accessible color palette generator

Discover beautiful color combinations your whole audience can appreciate and follow Web Content Accessibility Guidelines (WCAG) with ease.

[Random palette ↗](#)[#0073E6](#)[Generate](#)

The color pairings follow WCAG 2.1 AA based on a contrast ratio of 4.5:1. The pairings have sufficient contrast for use with normal text, large text and graphics.

Vibrant palette

#00BF7D	#0084C5	#0073E6	#2546F0	#592BED
Black Text Contrast: 8.79:1	Black Text Contrast: 8.39:1	White Text Contrast: 4.57:1	White Text Contrast: 6.54:1	White Text Contrast: 7.11:1

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Monochromatic palette

#B3C7F7	#8BABF1	#0073E6	#0461CF	#054FB9
Black Text Contrast: 12.43:1	Black Text Contrast: 9.19:1	White Text Contrast: 4.57:1	White Text Contrast: 5.51:1	White Text Contrast: 7.43:1

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Contrasting palette 1

#C44601	#F57600	#8BABF1	#0073E6	#054FB9
White Text Contrast: 4.97:1	Black Text Contrast: 7.47:1	Black Text Contrast: 9.39:1	White Text Contrast: 4.57:1	White Text Contrast: 7.43:1

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Accessible color palette generator

Discover beautiful color combinations your whole audience can appreciate and follow Web Content Accessibility Guidelines (WCAG) with ease.

[Random palette ↗](#)[#1005FB](#)[Generate](#)

The color pairings follow WCAG 2.1 AA based on a contrast ratio of 4.5:1. The pairings have sufficient contrast for use with normal text, large text and graphics.

Vibrant palette

#00A2F4	#025FFE	#1005FB	#8509EC	#F40FAF
Black Text Contrast: 7.67:1	White Text Contrast: 2.15:1	White Text Contrast: 6.62:1	White Text Contrast: 6.40:1	Black Text Contrast: 5.23:1

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Monochromatic palette

#B1B0FE	#8584FC	#1005FB	#0B03E0	#0B02C6
Black Text Contrast: 10.53:1	Black Text Contrast: 8.78:1	White Text Contrast: 6.62:1	White Text Contrast: 5.60:1	White Text Contrast: 11.60:1

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Contrasting palette 1

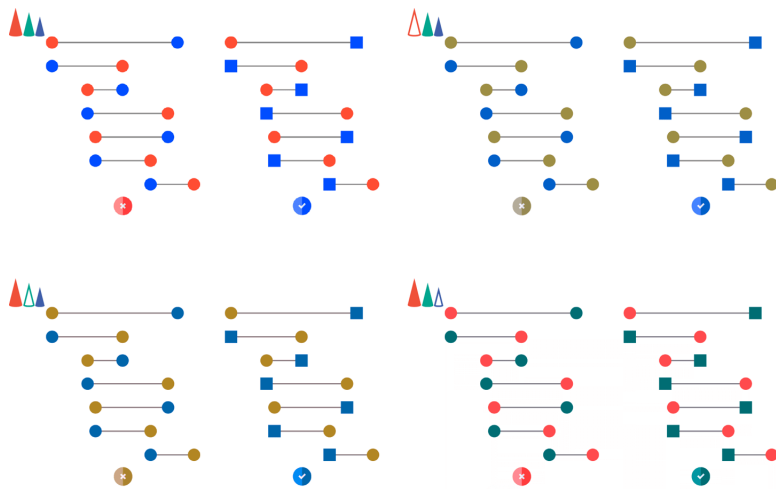
#9CB100	#BBDCC0	#8584FC	#1005FB	#0B02C6
Black Text Contrast: 5.61:1	Black Text Contrast: 13.16:1	Black Text Contrast: 6.30:1	White Text Contrast: 6.62:1	White Text Contrast: 11.60:1

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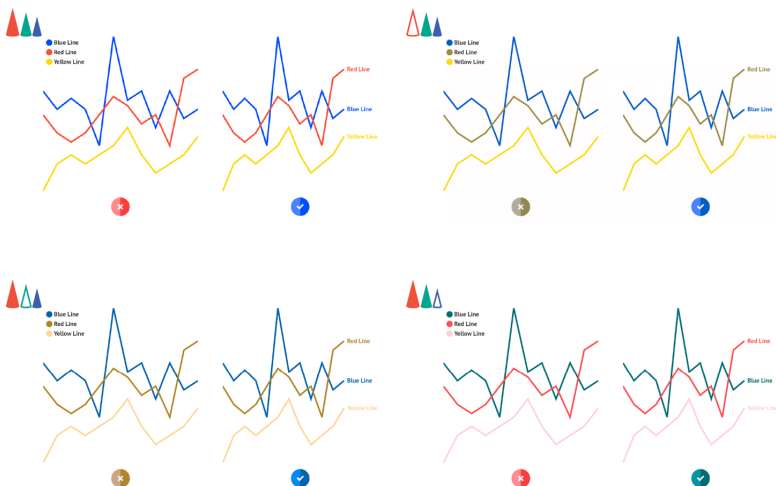


How to make charts color blind friendly?

Use shapes and icons as an addition or alternative to color-coding. In case one needs to use colors that are not perceptible by colorblind users, a chart can be saved by using icons as an addition to color, duplicating its informational function.

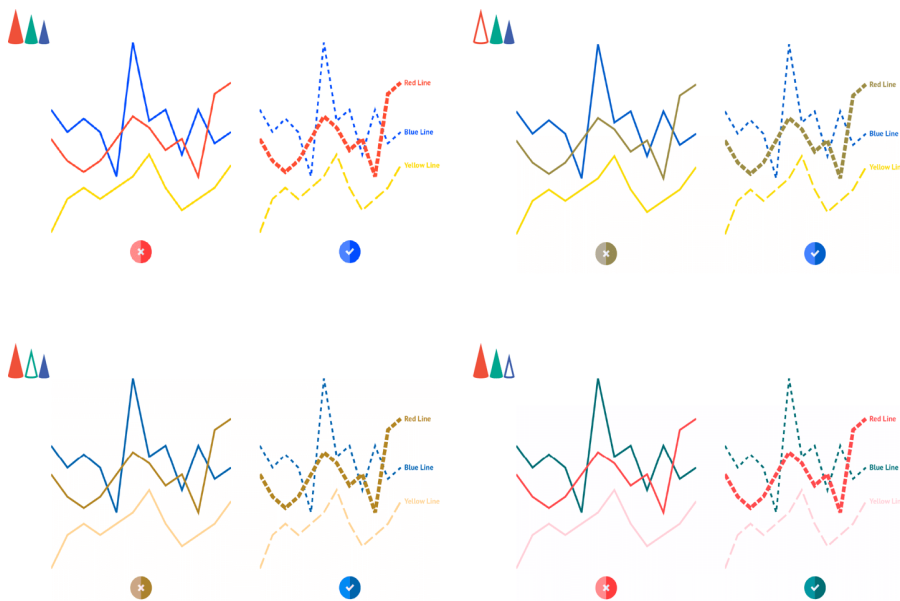


Typically, it's better to use direct labels instead of a legend – it saves the time and attention of a reader. Another advantage of direct labels is its ability to fix the usage of palettes that are not color blind friendly.

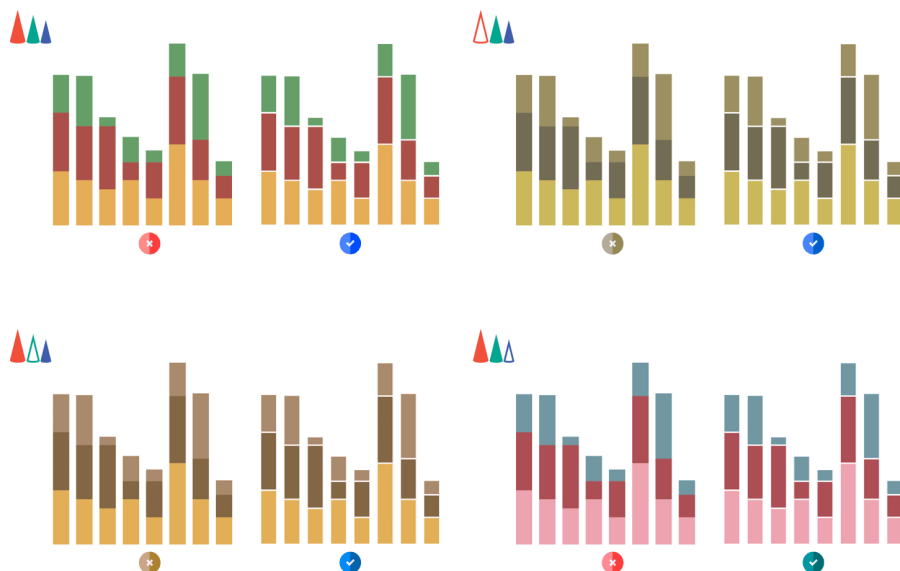


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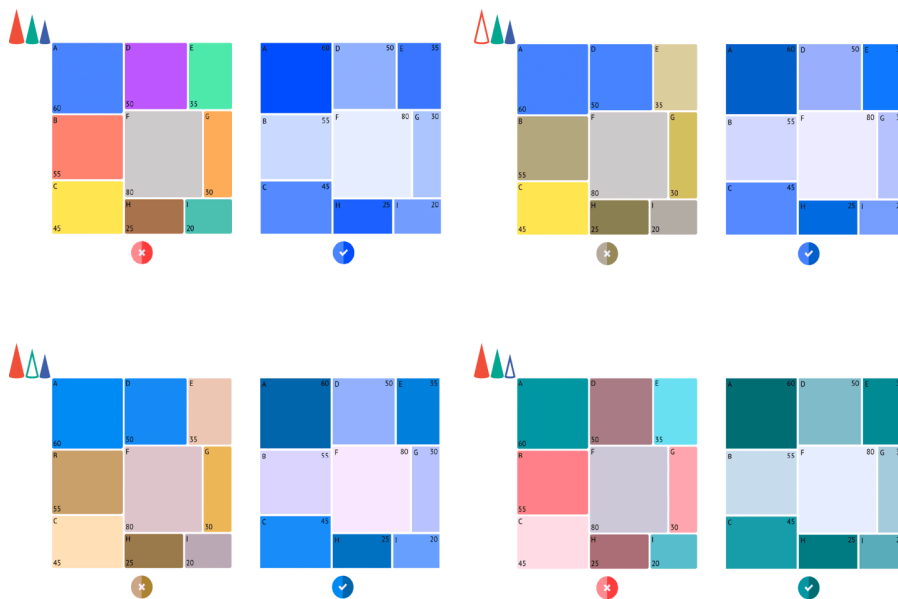
For line charts and their variations, dashed lines and lines with various stroke thicknesses can be a very helpful alternative to coloring.



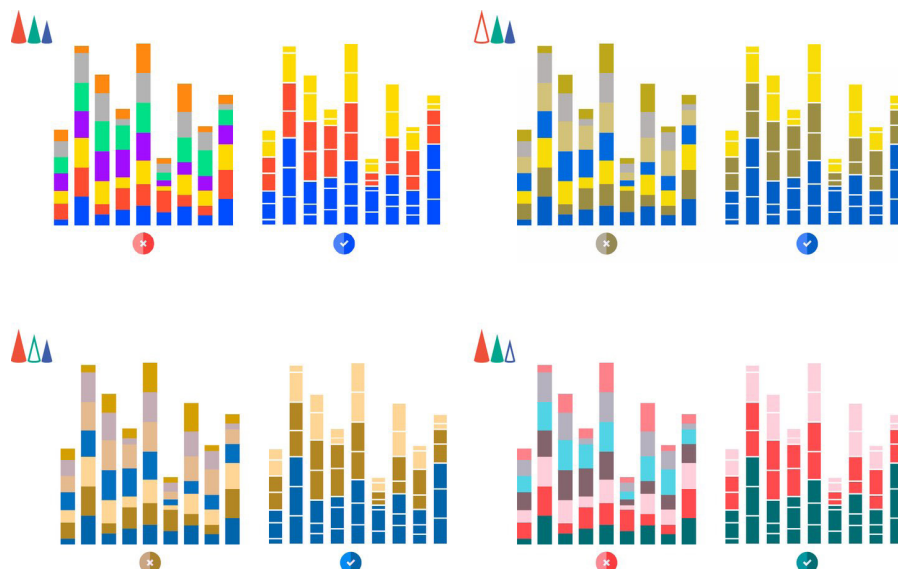
Adding strokes around chart elements might also help to distinguish one element from the other if the color might look the same for color-blind users.



If the usage of color is inevitable try to use a single hue palette. It will make the chart readable for all kinds of color blindness including monochromacy (when no color can be recognized). The second option is to use a red-yellow-blue palette. It will work for all kinds of color blindness except monochromacy.



Use color for groups, not the individual categories. By doing that the number of colors will be reduced, as well as visual clutter and the possibility of color confusion.



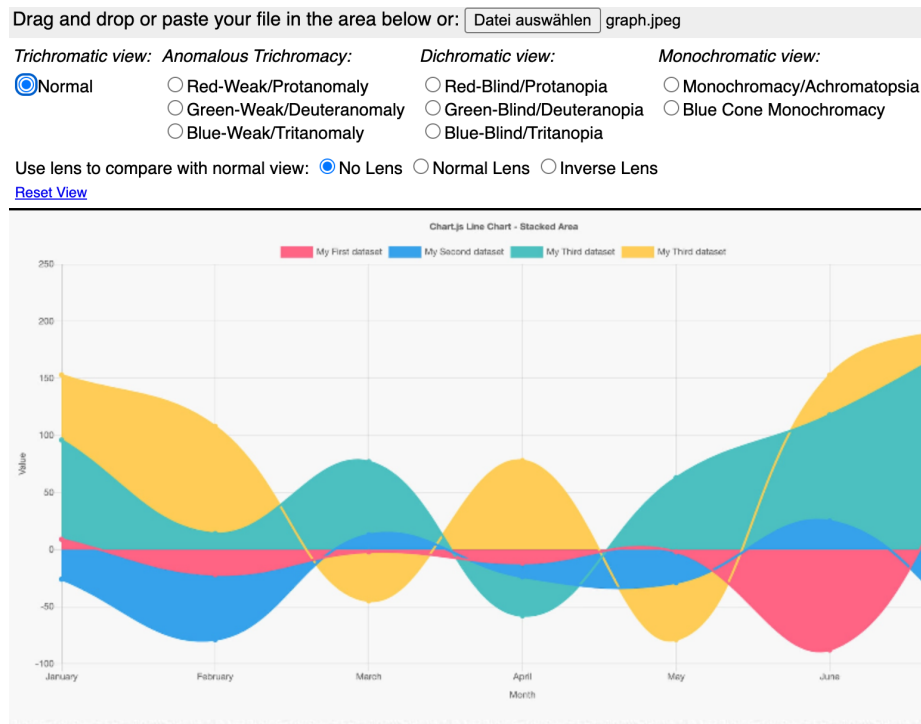
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Color Blindness Simulator

If you are not suffering from a color vision deficiency it is very hard to imagine how it looks like to be colorblind. The Color BLindness Simulator can close this gap for you. Just play around with it and get a feeling of how it is to have a color vision handicap.

As all the calculations are made on your local machine, no images are uploaded to the server. Therefore you can use images as big as you like, there are no restrictions. Be aware, there are some issues for the “Lens feature” on Edge and Internet Explorer. All others should support everything just fine.

Colblindor Color Blindness Simulator: www.color-blindness.com/coblis-color-blindness-simulator/



Drag and drop or paste your file in the area below or: graph.jpeg

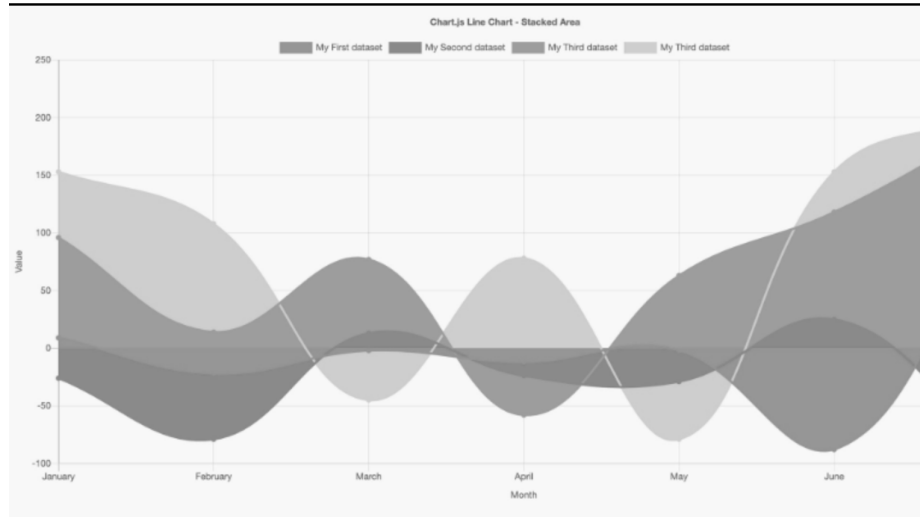
Trichromatic view: *Anomalous Trichromacy:*
☐ Normal ☐ Red-Weak/Protanomaly ☐ Green-Weak/Deuteranomaly ☐ Blue-Weak/Tritanomaly

Dichromatic view:
☐ Red-Blind/Protanopia ☐ Green-Blind/Deuteranopia ☐ Blue-Blind/Tritanopia

Monochromatic view:
☒ Monochromacy/Achromatopsia ☐ Blue Cone Monochromacy

Use lens to compare with normal view: ☒ No Lens ☐ Normal Lens ☐ Inverse Lens

[Reset View](#) [Open simulated image in new window](#)



Monochromacy/ Achromatopsia view

Drag and drop or paste your file in the area below or: graph.jpeg

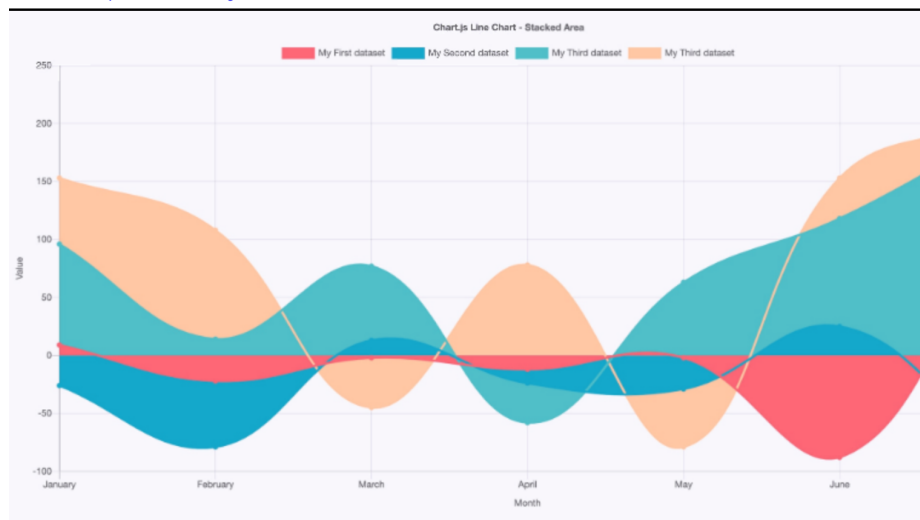
Trichromatic view: *Anomalous Trichromacy:*
☐ Normal ☐ Red-Weak/Protanomaly ☐ Green-Weak/Deuteranomaly ☒ Blue-Weak/Tritanomaly

Dichromatic view:
☐ Red-Blind/Protanopia ☐ Green-Blind/Deuteranopia ☐ Blue-Blind/Tritanopia

Monochromatic view:
☐ Monochromacy/Achromatopsia ☐ Blue Cone Monochromacy

Use lens to compare with normal view: ☒ No Lens ☐ Normal Lens ☐ Inverse Lens

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Blue-Weak/ Tritanomaly view

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Simulating Colorblindness in Photoshop

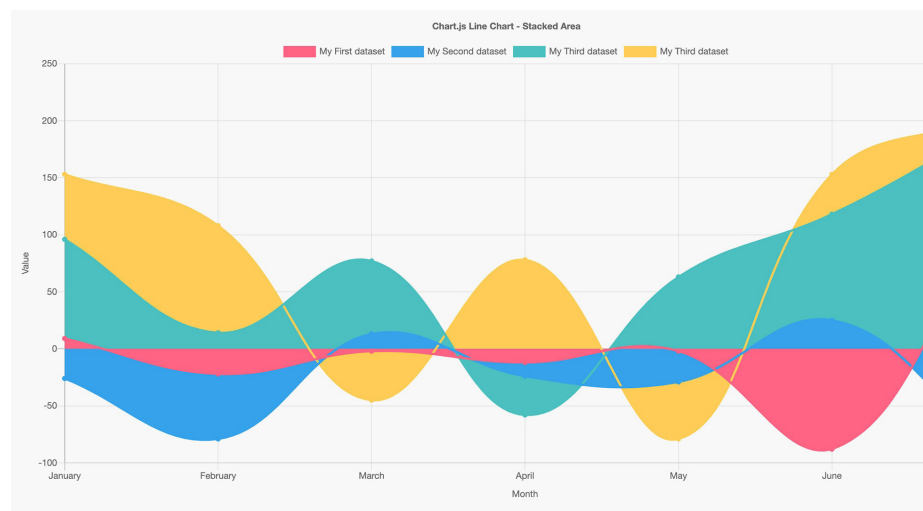
Photoshop has a built-in feature that can simulate what they'll see.

This tool is especially useful for graphic designers to make sure that designs are clear to the maximum number of people, but whether it's just out of curiosity or whether it's structuring the colors of a commercial advertising shoot, the tool can also be useful for photographers to get a sense of how different people are likely to perceive your image in terms of its colors.

It's very easy to do. And it does not affect the underlying image—think of it as a removable preview filter.

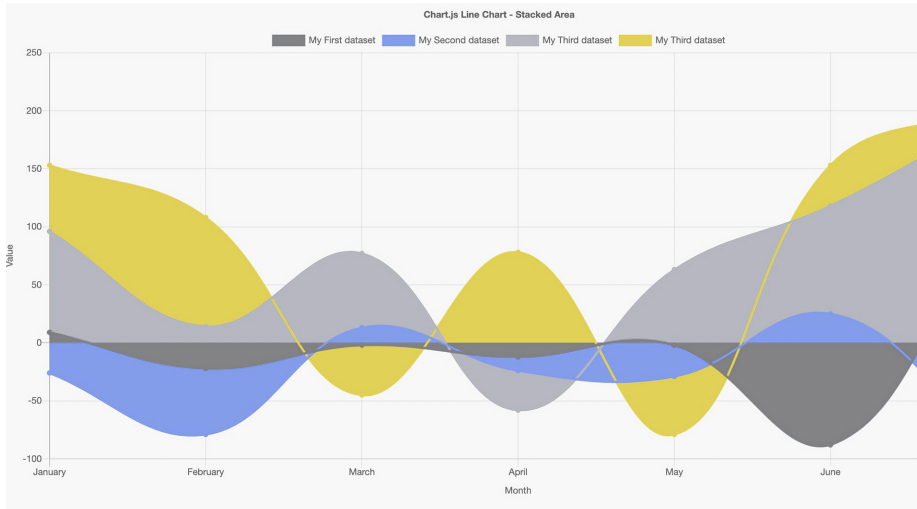
Open your image as you normally would.

Then go to View > Proof Setup. You have two choices: Protanopia-type or Deuteranopia-type. The effects are similar but not quite the same.

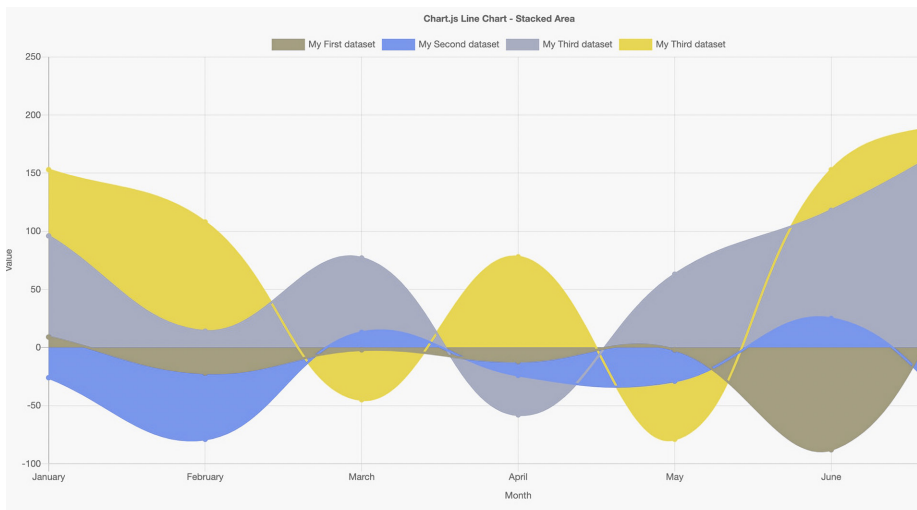


Normal view

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Protanopia view



Deuteranopia view

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Color combination examples

VARIANTE 1



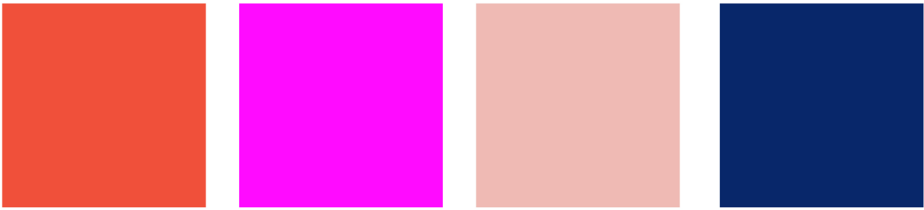
R: 233, G: 0, B: 6 R: 223, G: 0, B: 188 R: 10, G: 38, B: 255 R: 0, G: 22, B: 159



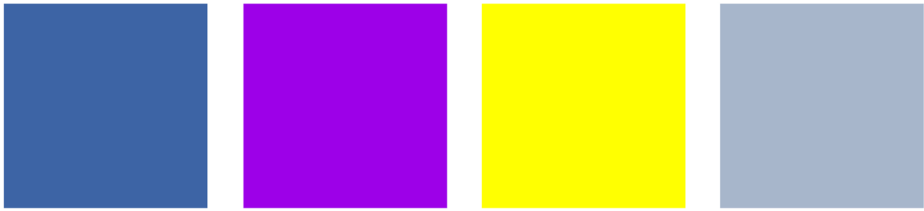
R: 220, G: 255, B: 180 R: 106, G: 0, B: 62 R: 255, G: 255, B: 0 R: 30, G: 203, B: 159

Normal view

VARIANTE 2



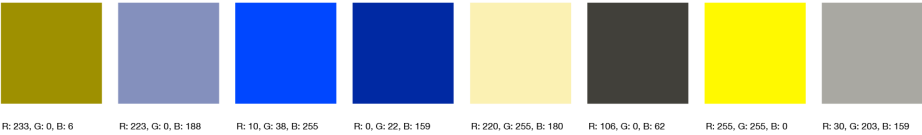
R: 240, G: 80, B: 57 R: 255, G: 11, B: 255 R: 238, G: 186, B: 180 R: 7, G: 38, B: 106



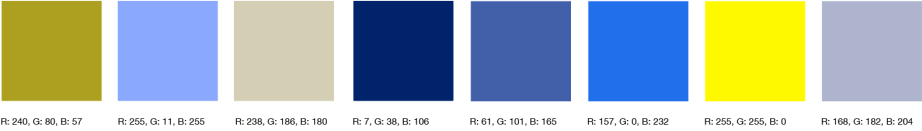
R: 61, G: 101, B: 165 R: 157, G: 0, B: 232 R: 255, G: 255, B: 0 R: 168, G: 182, B: 204

Normal view

VARIANTE 1

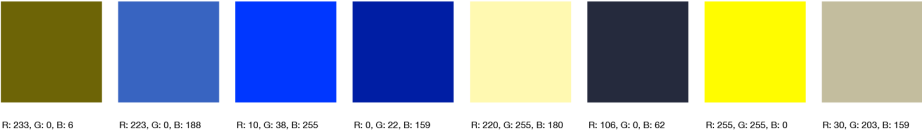


VARIANTE 2

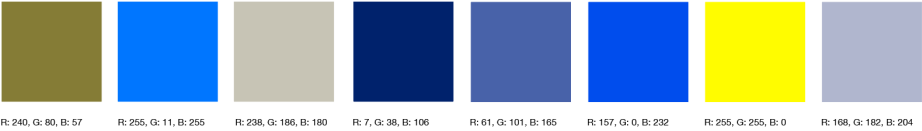


Deuteranopia view

VARIANTE 1



VARIANTE 2



Protanopia view

Sources

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<https://www.datylon.com/blog/data-visualization-for-colorblind-readers#:~:text=The%20first%20rule%20of%20making,out%20of%20these%20two%20hues.>

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